

Chao estimate other subfamilies

Load packages

```
library(dplyr)
library(SpadeR)
library(stringr)
estimator <- "    iChao1 (Chiu et al. 2014)"
```

List subfamilies from excel file

```
sub_families_list <- list.files("../Data/Bold_only/BIN_abundance_subfamilies/",
                                full.names = TRUE)
```

Calculate iChao1 for both reared and core and compile for all species

```
sub_family_outputs_all <- data.frame()

for(fm in sub_families_list){

  ## load data

  data_all <- read.csv(fm, skip = 1)

  ## filter for core and reared

  core <- select(data_all, BIN, count) %>%
    filter(!is.na(count)) %>%
    mutate(source = 'core')

  reared <- select(data_all, BIN= BIN.1, count = count.1) %>%
    filter(!is.na(count)) %>%
```

```

mutate(source = 'reared')

## summarise total bins

total <-
  bind_rows(core, reared) %>%
  group_by(BIN) %>%
  summarise(count = sum(count))

## calculate shared, reared only and maialse only

shared <- unique(c(core$BIN[core$BIN %in% reared$BIN],
  reared$BIN[reared$BIN %in% core$BIN]))

reared_only <- reared$BIN[!reared$BIN %in% core$BIN]

core_only <- core$BIN[!core$BIN %in% reared$BIN]

## Estimate iChao for core -- use try catch for subfamilies with too few bins

core_estimate <- tryCatch({
  ChaoSpecies(core$count, "abundance", k=10, conf=0.95)},
  error = function(e) {
    template_result <- ChaoSpecies(reared$count, "abundance", k=10, conf=0.95)
    na_matrix <- template_result
    na_matrix$Species_table[] <- NA
    na_matrix$Basic_data_information[] <- NA
    return(na_matrix)
  })

## Estimate iChao for reared

reared_estimate <- ChaoSpecies(reared$count, "abundance", k=10, conf=0.95)

subfamily <- str_remove(str_remove(str_extract(fm, "[A-Z][a-z]*-"), "/"), "-")

## compile results in data frame

sub_family_outputs_1 <- data.frame(subfamily = subfamily,
  total_n_bin = nrow(total),
  shared_bins = length(shared),
  proportion_shared = length(shared)/nrow(total),

```

```

nreared_total_bin = nrow(reared),
reared_only = length(reared_only),
core_only = length(core_only),
reared_Chao1_estimate = reared_estimate$Species_table[estimator, nreared_total_bin],
reared_Chao1_se = reared_estimate$Species_table[estimator, nreared_total_bin + 1],
reared_Chao1_95_lower = reared_estimate$Species_table[estimator, nreared_total_bin + 2],
reared_Chao1_95_upper = reared_estimate$Species_table[estimator, nreared_total_bin + 3],
reared_Chao1_CV = reared_estimate$Basic_data_information[nreared_total_bin, estimator],
reared_Chao1_Coverage = reared_estimate$Basic_data_information[nreared_total_bin + 4, estimator],
core_total_bin = nrow(core), core_Chao1_estimate = core_estimate$Species_table[estimator, core_total_bin],
core_Chao1_se = core_estimate$Species_table[estimator, core_total_bin + 1],
core_Chao1_95_lower = core_estimate$Species_table[estimator, core_total_bin + 2],
core_Chao1_95_upper = core_estimate$Species_table[estimator, core_total_bin + 3],
core_Chao1_CV = core_estimate$Basic_data_information[core_total_bin, estimator],
core_Chao1_Coverage = core_estimate$Basic_data_information[core_total_bin + 4, estimator],

sub_family_outputs_all <- bind_rows(sub_family_outputs_all, sub_family_outputs_1)
}

```

Warning: In this case, it can't estimate the variance of 2nd-order-jackknife estimation

Warning: In this case, it can't estimate the variance of 2nd-order-jackknife estimation

Warning: In this case, it can't estimate the variance of Homogeneous estimation

```
rownames(sub_family_outputs_all) <- NULL
```

```
sub_family_outputs_all
```

	subfamily	total_n_bin	shared_bins	proportion_shared	nreared_total_bin
1	Agathidinae	150	23	0.15333333	141
2	Anomaloninae	50	4	0.08000000	44
3	Campopleginae	333	71	0.21321321	243
4	Cardiochilinae	13	1	0.07692308	12
5	Cheloninae	190	4	0.02105263	77
6	Euphorinae	186	8	0.04301075	58
7	Homolobinae	10	0	0.00000000	8
8	Macrocentrinae	57	12	0.21052632	43
9	Mesochorinae	185	15	0.08108108	152
10	Metopiinae	88	5	0.05681818	63
11	Microgastrinae	1165	109	0.09356223	886

12	Ophioninae	138	18	0.13043478	133
13	Orgilinae	55	3	0.05454545	28
14	Rogadinae	168	9	0.05357143	98
15	Tryphoninae	54	10	0.18518519	43
	reared_only	core_only	reared_Chao1_estimate	reared_Chao1_se	
1	118	9	168.064	6.473	
2	40	6	59.032	7.766	
3	172	90	304.507	10.447	
4	11	1	12.000	0.728	
5	73	113	123.843	13.300	
6	50	128	77.698	6.813	
7	8	2	19.128	9.194	
8	31	14	137.553	71.819	
9	137	33	218.738	13.626	
10	58	25	86.618	10.156	
11	777	279	1134.972	19.763	
12	115	5	206.972	18.886	
13	25	27	32.354	1.884	
14	89	70	157.254	15.041	
15	33	11	99.451	27.510	
	reared_Chao1_95_lower	reared_Chao1_95_upper	reared_Chao1_CV		
1	158.046	183.970	1.428		
2	49.795	82.996	1.388		
3	287.195	328.601	2.546		
4	12.000	14.171	0.749		
5	104.141	157.846	2.779		
6	68.193	96.066	1.74		
7	10.706	53.756	2.284		
8	68.189	397.934	1.966		
9	196.911	251.171	1.619		
10	73.535	115.951	2.031		
11	1099.153	1176.811	2.543		
12	178.202	254.052	1.879		
13	29.933	37.806	1.102		
14	134.309	194.699	1.753		
15	65.843	182.505	1.143		
	reared_Chao1_Coverage	core_total_bin	core_Chao1_estimate	core_Chao1_se	
1	0.979	32	45.686	5.595	
2	0.953	10	17.934	7.401	
3	0.986	161	269.703	22.203	
4	1	2	NA	NA	
5	0.968	117	277.920	41.817	
6	0.975	136	212.718	14.485	

7	0.939	2	2.000	0.477
8	0.959	26	56.823	18.965
9	0.93	48	83.215	10.334
10	0.98	30	81.953	32.096
11	0.98	388	724.922	46.392
12	0.957	23	42.306	12.567
13	0.959	30	50.908	7.778
14	0.883	79	122.162	12.389
15	0.822	21	63.058	32.775
	core_Chao1_95_lower	core_Chao1_95_upper	core_Chao1_CV	core_Chao1_Coverage
1	38.333	61.572	2.159	0.927
2	11.683	47.409	1.904	0.956
3	234.140	322.557	4.204	0.966
4	NA	NA	<NA>	<NA>
5	214.503	382.583	3.088	0.947
6	189.161	246.714	2.163	0.922
7	2.000	3.597	0.964	1
8	36.153	119.578	2.049	0.9
9	68.048	109.856	1.129	0.711
10	47.046	188.342	4.568	0.969
11	645.557	828.743	3.39	0.953
12	29.022	84.892	1.472	0.833
13	40.324	72.341	1.359	0.91
14	103.866	153.921	2.314	0.942
15	31.901	183.259	0.978	0.771